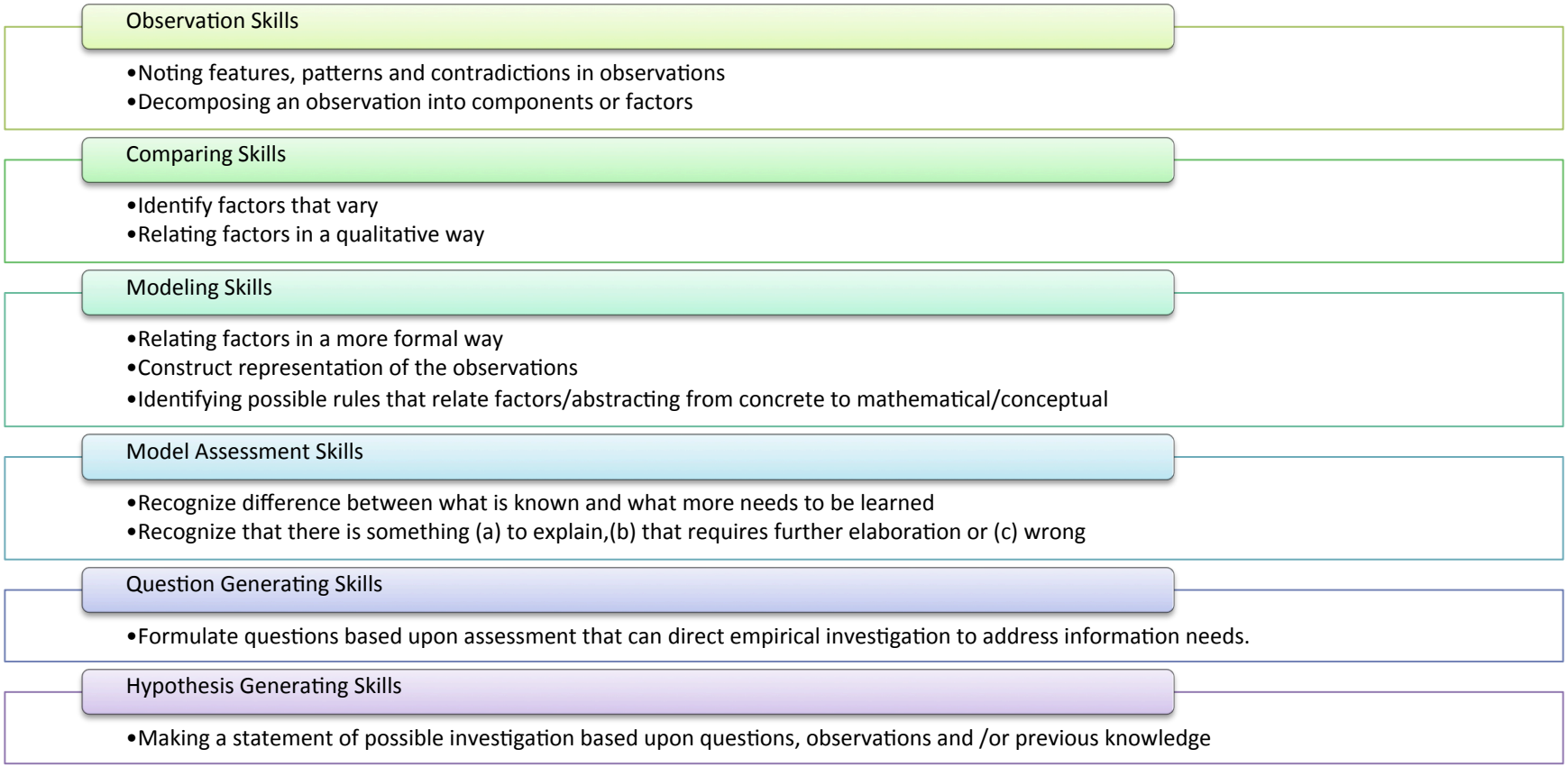
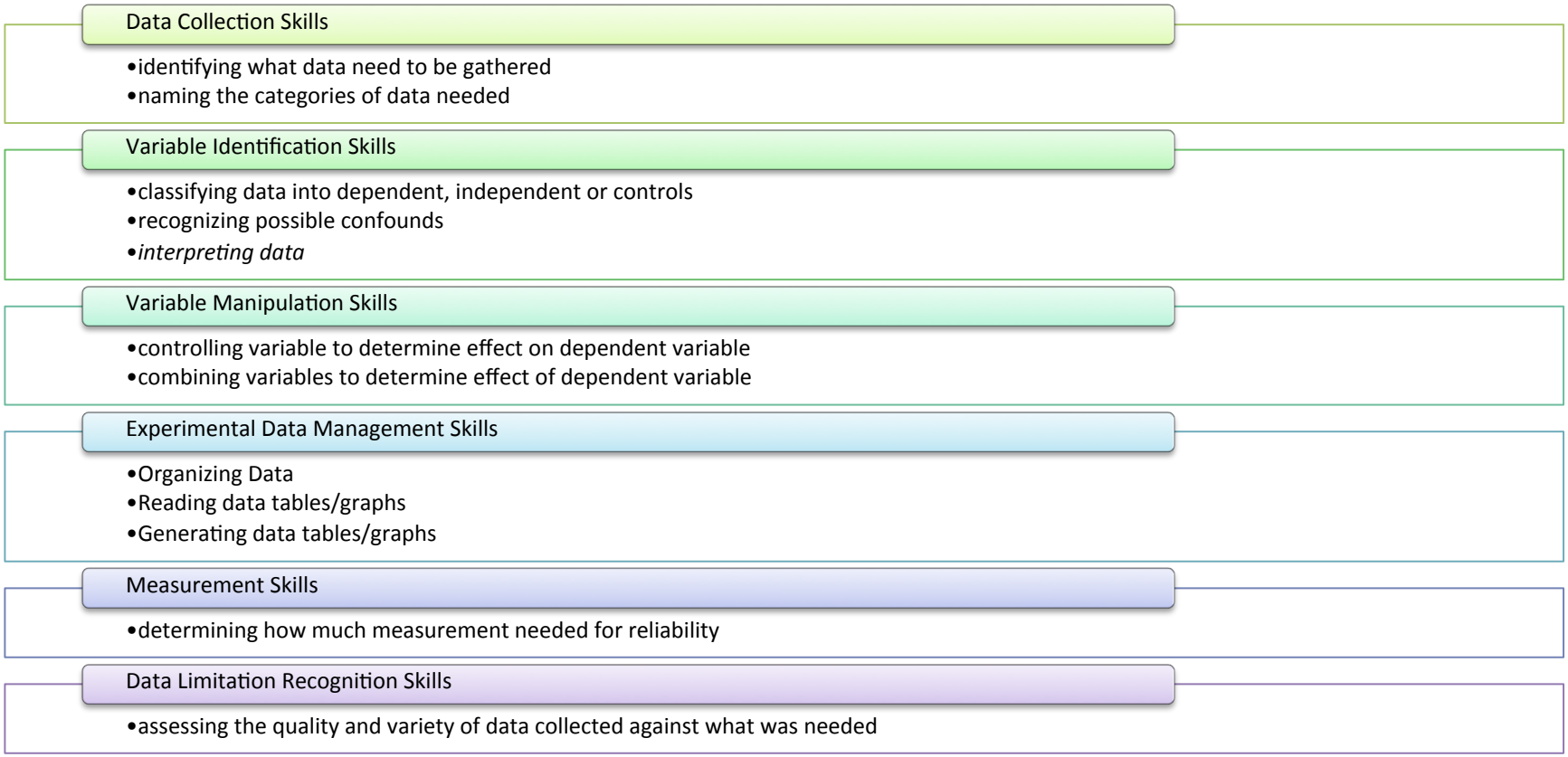
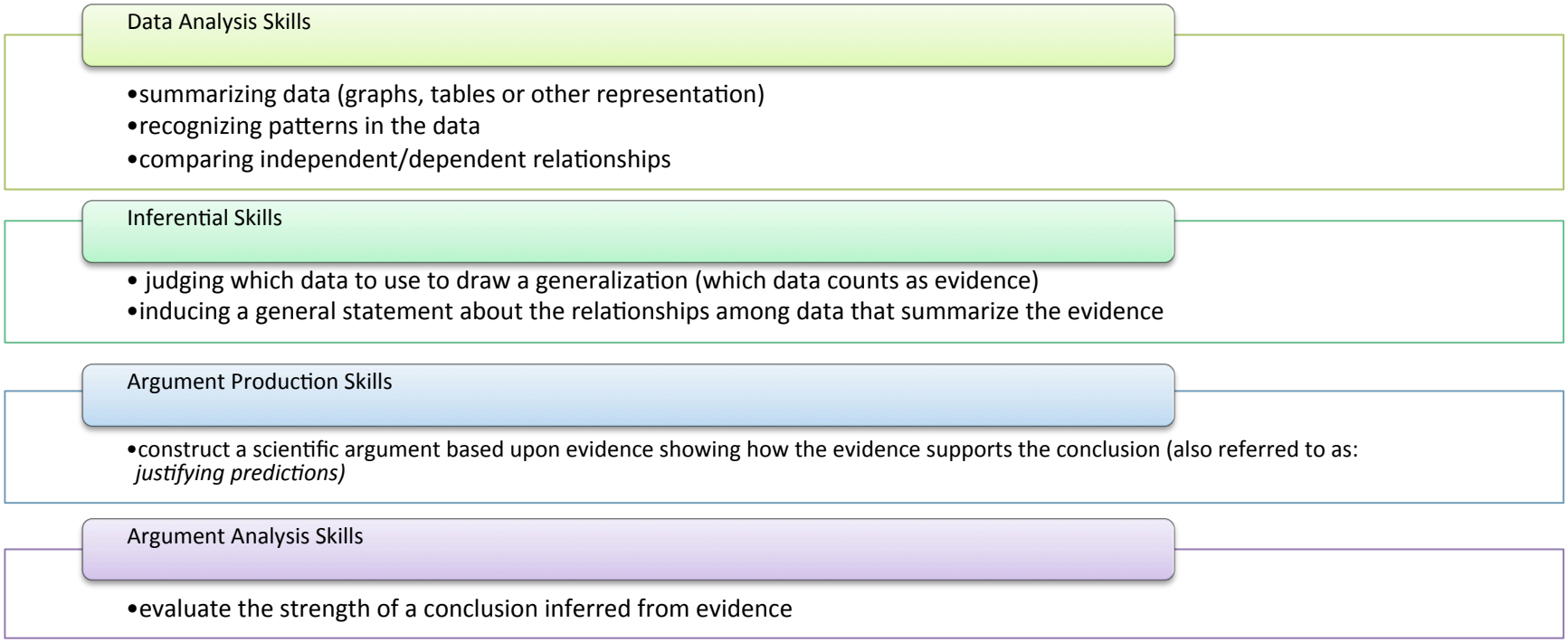
Facet 1: **Generating a hypothesis to account for observations** – the ability to observe a situation or event, recognize that there is something to find out, recognize the difference between existing understanding and what more needs to be learned and to clearly articulate a question that can direct an empirical investigation.



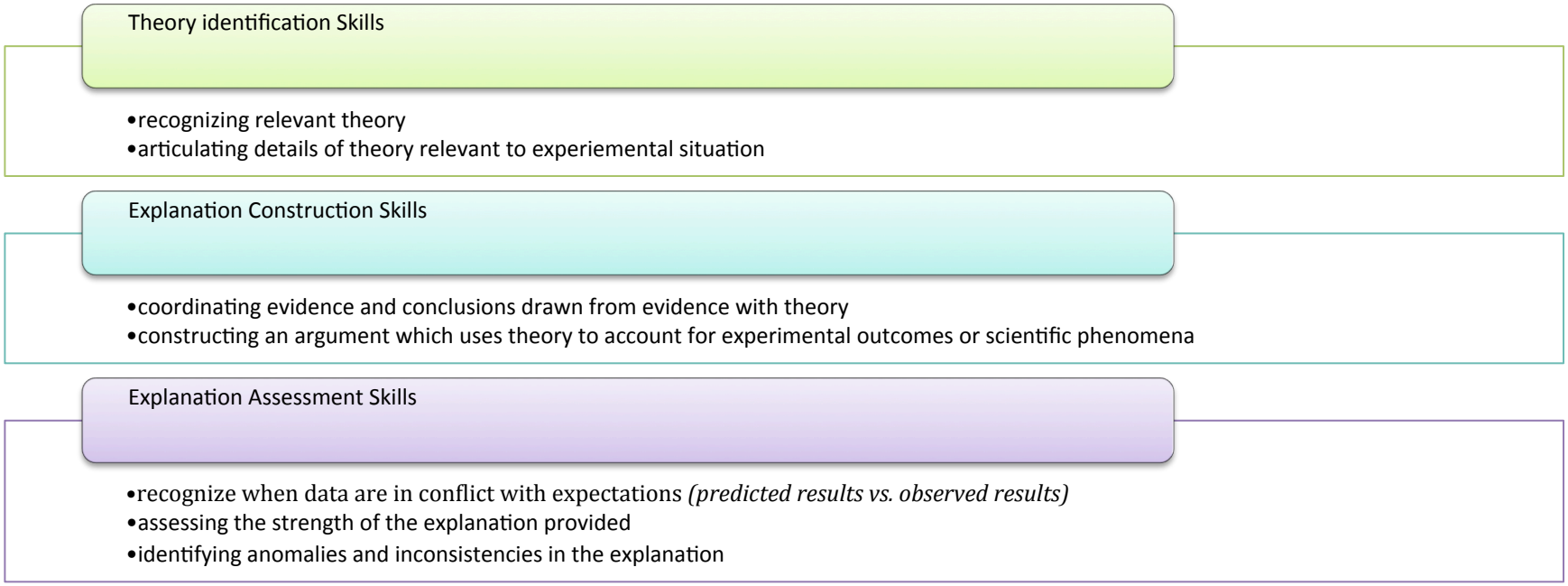
Facet 2: **Hypothesis Testing** – the ability to *design tests* of a hypothesis that correctly *identify* and *manipulate* all relevant *variables* in order that empirical evidence may be produced that will allow one to answer questions.



Facet 3: **Reasoning from evidence to conclusions** – the ability to interpret the results of an investigation and to draw justified inferences/conclusions based upon that data.



Facet 4: **Generating explanations** - the ability to coordinate theory and evidence in such a way so as to draw inferences that account for either causal relationships or stochastic relationships. These activities employ the theory, seek underlying theoretical causes for the evidence and utilize models to describe patterns in the data or scientific phenomena.



Facet 5: **Reconciling theory and evidence (RTE)** - the ability to evaluate theory in light of experimental conclusion, reconcile new evidence with prior beliefs, and (if required) revise one's theory and generate new predictions.

